

# Effectiveness of a web-based intervention for individuals who committed sexual offenses against children: a concealed, prospective, parallel group, placebo-controlled, randomized trial in Germany.

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## Summary

### *Background*

Sexual offenders often present with diverse risk profiles requiring differentiated interventions<sup>?</sup>.

### *Methods*

We conducted a randomized controlled trial evaluating a web-based relapse prevention tool. The trial was prospectively registered on April 24, 2020 in the German Clinical Trial Register (DRKS00021256).

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### *Findings*

Preliminary findings suggest significant reductions in dynamic risk-factors.

### *Interpretations*

Taken together, the intervention did not lead to measurable changes in the primary CARES outcome but was associated with selected improvements in secondary endpoints — most notably reduced explicit sexual interest and strengthened working alliance with coaches and supervisors — without indications of harm also for low-risk individuals.

### *Funding*

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## Research in context

### *Evidence before this study*

Before undertaking this study, a XXX search revealed no published results on the efficacy of web-based intervention in reducing risk-factors for recidivism of Individuals who committed CSA or CSEM Offences (ICCO). The results of these reviews have been published online 2021. In summary, there existed no study reporting on the effectiveness of WBIs. The results of a placebo-controlled randomized trial of an WBI exclusively for CSEM users in the dark field have been reported for the first time in 2022. The RCT demonstrated a small, but significant reduction of time spend on viewing CSEM based on self-reports. In 2024, an updated review revealed no further empirical evidence for the effectiveness of WBIs. More recent meta-analyses did not provide additional empirical evidence for WBIs. To the best of our knowledge, at the moment there is no evidence for the effectiveness of WBIS in reducing re-offenses of CSA and CSEM.

### *Added value of this study*

The study provides for the first time data on the effectiveness of a WBI for individuals who have already committed CSA or CSEM offences. While the main outcome revealed no superiority of the WBI against a placebo condition, secondary outcomes provide hints for superiority of the intervention arm in reducing sexual interest. The WBI significantly enhanced the therapeutic alliance with online coaches and community supervisors only in the intervention arm. Adverse events were low as well as re-offenses during trial participation.

### *Implications of all the available evidence*

Authors should state the implications for practice or policy and future research of their study combined with existing evidence.

## 1. Introduction

Child Sexual Abuse (CSA) constitutes one of the most severe adverse experiences in child development.<sup>1</sup> The creation, sharing, and viewing of Child Sexual Exploitation Material (CSEM) increased significantly in the last years<sup>2</sup>. Beyond its profound short- and long-term consequences for survivors, CSA represents a substantial challenge for society due to high economical costs<sup>3</sup>. These figures clearly demonstrate the importance of preventing CSA and CSEM.

In CSA and CSEM prevention, a distinction is commonly drawn between primary, secondary, and tertiary prevention programs. Primary prevention initiatives are directed at children and their parents, whereas secondary prevention programs seek to engage adults identified as being at risk for sexual child offenses, but have not yet been convicted. Tertiary programs have the longest history and are designed for Individuals who already committed CSA or CSEM Offenses (ICCO).<sup>3</sup> In recent years, more and more secondary prevention programs for ICCO have been implemented.<sup>4</sup> However, a recent meta-analysis identified only five publications exploring the effectiveness of secondary prevention for ICCO, none of them with a methodological high quality.<sup>3,4</sup> Thus, currently it is not possible to draw meaningful conclusions on the effectiveness of secondary treatment programs.<sup>4</sup> The effectiveness of tertiary treatment programs for ICCO has been studied more extensively. In a recent meta-meta analysis, including 21 meta-analyses on tertiary treatment programs, a overall treatment effect with an odds ratio (OR) of 0.67 was reported.<sup>5</sup> Several moderators have been identified: community treatment programs provide higher effect-sizes than institutional programs, individualized programs show higher effect-sizes than group programs, high-intensity programs are more effective than low-intensity programs, and the higher the methodological quality, the lower the effect sizes.<sup>6,5</sup> Up to now, most of the meta-analyses report on tertiary programs for individuals who have been convicted for CSA. The figure for CSEM is not as well studied. A recent review demonstrated preliminary support for the effectiveness of these programs, but also shed light on the methodological weakness of existing studies on the effectiveness of prevention programs for individuals who have used CSEM: mostly, they suffer from small sample sizes, too short follow-up periods and a lack of officially registered re-offenses as outcome measure.<sup>7</sup> To sum up, there exists a broad empirical basis for a small but significant effectsize in reducing recidivism by tertiary programs. For individuals who have been sentenced for CSEM as well as for secondary prevention programs, the empirical proof of effectiveness is still pending.

Originating from only having few possibilities of community-based treatment offers (e.g. the majority of outpatient therapists working in the primary healthcare system are unwilling to provide treatment) and a rising number of CSEM offenders, there exists a post-release care gap for ICCO.<sup>8,2</sup> WBIs are assumed to provide greater accessibility for individuals who face challenges in availing mental health services.<sup>9</sup> Thus, it was assumed that WBIs may bridge the gap in the care of ICCO, but empirical evaluations are scarce.<sup>10,11,12</sup> Schröder et al. identified six WBIs implemented in secondary prevention without published data on their effectiveness.<sup>12</sup> Up to now, only one WBI (Prevent It<sup>13</sup>) reported on it's effectiveness. The authors evaluated a guided WBI on a sample of 160 individuals who viewed more than five hours weekly CSEM at baseline<sup>14</sup> The Randomized Controlled Trial (RCT) showed, that, compared with a psychological placebo, participants

of the intervention arm reported a significant, but small decrease in weekly viewing of CSEM (Cohen's  $d = 0.18$ ). Besides a high dropout rate, this study provides first evidence for the effectiveness of WBIs for secondary prevention of individuals viewing CSEM. Only a minority of the sample reported prior convictions (6.9%).<sup>11,4</sup> Data on an updated version are not presented yet.<sup>15</sup> Thus, for ICCO, to the best of our knowledge, there exist no proofed WBI yet.

### *1.1. Aim of the study*

The study aimed to evaluate the effectiveness of a newly developed WBI @myTabu for ICCO compared to a psychological placebo in a RCT. The primary hypothesis assumes a significant reduction of dynamic risk-factors from baseline to inter-, and post-treatment, compared to a placebo arm. It was further assumed, that each online module within the intervention arm will reduce the dynamic risk-factors targeted in this module in a statistically significant manner.

## **2. Method**

### *2.1. Study design and participants*

The trial was conducted online in Germany and follows a concealed, prospective, parallel group, placebo-control trial with 1:1 randomization. The study design has been prospectively published.<sup>16</sup> The trial was prospectively registered on April 24, 2020 in the [German Clinical Trial Register \(DRKS00021256\)](#). The research project was advised by a multidisciplinary team including a representative of participants. All individuals under community supervision (CS; §§56, 57, 68 StGB) from all German federal states (except Hamburg, Bremen, and Saarland), which are sentenced for at least one case of CSA (§§176ff StGB) or CSEM (§184b StGB), were eligible. Exclusion criteria were younger than age of 18 years, a probation period shorter than six months, no access to online intervention, a severe acute psychiatric or cerebro-organic disorder, or a severe cognitive impairment. All participants gave written informed consent. The participants were credited with monetary compensation (€120). However, the participants did not receive any confirmation of having met the treatment requirements set by court. Ethics approval was granted by the medical ethical board of the University Medical Center Göttingen, Göttingen, Germany, on February 21, 2020 (reference number 16/2/20).

### *2.2. Randomisation and masking*

The trial comprises two parallel arms (intervention arm and placebo arm). Subjects are randomly allocated to treatment using an allocation ratio of 1:1. The randomization lists are centrally generated using a computerized system (secuTrial®, interActive Systems GmbH, Germany), stratified by the offense type (§176ff StGB vs. §184b StGB), type of CS, external treatment as ordered by court, and static risk to re-offend at baseline (assessed with the modified Static-99<sup>17</sup>). At screening, each subject receives the next consecutive screening number. At randomization, each subject eligible for study participation receives the next consecutive randomization number according to his stratum from a block of randomization numbers (block size: 4). Due to security reasons, blinding of Supervision Officers (SOs) and Study Therapists (STs) was not possible. Nevertheless, participants, outcome adjudicators, and data analysts were blinded until the end of the data analysis.

### *2.3. Procedures*

The concept of the WBI framework, theoretical foundation, and content of the WBI are published online<sup>16</sup> and shown in the appendix Section 2 and Figure 1. The content of the trial arms are summarized in Table 1. The WBI builds on already established in-face community-based treatment programs following a cognitive-behavioral approach.<sup>18,19</sup> The intervention arm focus solely on psychological meaningful risk-factors.<sup>20</sup>

Table 1: Detailed description of the WBI content.

| Module          | Intervention Content                                                                                                                                                                                                                                                                 | Placebo Content                                                                                                                                                                                                                                             |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Module 1</b> | <ul style="list-style-type: none"> <li>• Identification of personal values and long-term goals to enhance treatment readiness.</li> <li>• Structured decision-making and goal-setting to prioritize long-term outcomes over immediate rewards.</li> </ul>                            | <ul style="list-style-type: none"> <li>• Psychoeducation on benefits of physical activity and reduction of sedentary behavior.</li> <li>• Goal-setting to increase motivation and support sustainable active lifestyles.</li> </ul>                         |
| <b>Module 2</b> | <ul style="list-style-type: none"> <li>• Development of collaborative skills to improve engagement and adherence within community supervision.</li> <li>• Cognitive-behavioral strategies addressing trauma-related avoidance and relational functioning.</li> </ul>                 | <ul style="list-style-type: none"> <li>• Education on muscle function and physical capabilities.</li> <li>• Identification of motivational factors and social influences on training behavior.</li> </ul>                                                   |
| <b>Module 3</b> | <ul style="list-style-type: none"> <li>• Linking emotions to unmet needs to reduce impulsive responding.</li> <li>• Mindfulness-based strategies to improve emotional awareness and self-regulation.</li> </ul>                                                                      | <ul style="list-style-type: none"> <li>• Education on sleep quality, stress, and emotional regulation.</li> <li>• Training in behavioral strategies to stabilize sleep routines.</li> </ul>                                                                 |
| <b>Module 4</b> | <ul style="list-style-type: none"> <li>• Structured problem-solving training to address self-regulation deficits.</li> <li>• Case-based application to improve decision-making and solution evaluation.</li> </ul>                                                                   | <ul style="list-style-type: none"> <li>• Education on dream physiology and stress-related sleep disruption.</li> <li>• Reflective exercises linking sleep, stress, and executive functioning.</li> </ul>                                                    |
| <b>Module 5</b> | <ul style="list-style-type: none"> <li>• Identification and restructuring of offense-supportive cognitive distortions.</li> <li>• Case-based illustration of consequences of distorted thinking.</li> <li>• Need principle: choose three out of 16 pre-defined cognitions</li> </ul> | <ul style="list-style-type: none"> <li>• Education on nutrition and the mind-body connection.</li> <li>• Practical skills to support sustainable dietary change.</li> <li>• adaptive: learn more on three out of six pre-defined nutrition types</li> </ul> |
| <b>Module 6</b> | <ul style="list-style-type: none"> <li>• Differentiation of problematic and healthy sexual interests.</li> <li>• Cognitive and behavioral strategies to manage high-risk triggers.</li> </ul>                                                                                        | <ul style="list-style-type: none"> <li>• Conceptualization of psychological well-being beyond happiness.</li> <li>• Self-care practices linking nutrition, bodily awareness, and mental stability.</li> </ul>                                               |

For the recruitment, scientific staff informed SOs about the trial and asked to inform individuals under their supervision. If interested, a meeting with a research assistant (RA) took place at the supervision office. After obtaining the written informed consent, the RA extracted data from official records on site. The WBI was divided into six modules, each module was further divided into four sessions. Participants were advised to walk through one session per week and finish one module in one month. The next session becomes unlocked, when one week has past and all guided exercises have been successfully completed. User tests before running the trial took approximately 60 minutes to complete one session. All endpoints were provided online (see appendix Table 1). Due to ethical reasons, before presenting the questionnaire forms for the primary outcome and the Mood and Risk Questionnaire (MRQ), participants were informed that these answers will be viewable by their SO. To exclude a participant, the SO had to fill in a questionnaire on exclusion reasons (Exclusion Questionnaire, EQ). There was no time constraint defined at which a participant was excluded. Due to data security, the trial staff had no possibility to contact a participant directly. Communication between STs and participants took place online. STs (one male psychotherapist, four female psychologists) were intensively trained by completing a practice online treatment of at least one simulated participant. Regular monthly group supervisions ensured treatment consistency; the answers of STs on guided exercises were manualized (appendix Figure 2). To minimize therapist-related effects, the same STs supported both trial arms.

#### 2.4. Outcomes

References, psychometric criteria and descriptions for all outcomes are provided in the appendix Section 3. The Statistical Analysis Plan (SAP), defines officially recorded re-offenses five years after last-patient-out (at September 2030) and the Index of Desistance (IoD) as primary endpoints.<sup>16</sup> However, recently, the items of the scales underlying the IoD have been reduced and one scale was removed due to low internal consistency; the IoD was renamed into Child Sexual Abuse Risk Evaluation Self-Report (CARES). The CARES is a composite measure with higher scores indicating greater severity of both acute and stable dynamic risk-factors.<sup>21</sup> We report the validated CARES (see appendix section 6.2 for pre-specified IoD).

Secondary endpoints include specific questionnaires operationalizing these risk-factors that are thematically addressed in the individual modules (appendix Table ??).

Frequencies of re-offenses as reported by SOs were assessed with the EQ. Frequencies of critical events as reported by STs were assessed continuous (appendix Figure 4), adverse events with the Mood and Risk Questionnaire (MRQ), post-hoc defined as the occurrence of at least one report of the highest possible score on a sub-scale.

The Working Alliance Inventory-Short Revised (WAI-SR) was used to assess therapeutic alliance with STs and SOs. Changes in psychological well-being have been assessed with the WHO-5 Well-Being Index (WHO-5). To assess a possible volunteer-effect, the Questionnaire of Non-Participation (QNP) was developed. The Questionnaire for Acceptance of Technology (ATQ) and the Questionnaire of Subjective Therapy Preconditions (SBV-R) were administered, but not related to the hypotheses and not reported here. Preliminary ATQ data from the intermediate analysis are published.<sup>22,23</sup>



### 2.5. Statistical analysis

Data analysis was performed using the software R.<sup>24</sup> Interim analysis was done by a independent data safety monitoring board (DSMB) on April 20, 2022. Stopping guidelines have been prospectively defined (appendix Section 1).

We found no published WBI for ICCO to guide power calculation, and conservatively assumed small effects. Power analysis revealed a minimum of 214 participants in each arm to be able to detect small effects ( $d = 0.27$ ) with a power of 80% at a significance level  $\alpha = 5\%$ .

Time participants needed to finish the WBI (the time from beginning of module one to the end of the module) and working time (the time, a participant was working on the content of a module) was analyzed by means of Gaussian linear model for repeated measures (MMRM) with treatment group, time (six time-points), and treatment-by-time interaction.

Statistical analysis for the primary outcome follows the intention-to-treat principle (ITT). The ITT population was defined as all participants who have completed at least one module. The CARES was analyzed by MMRM with treatment group, time (six time-points), treatment-by-time interaction, and the randomization strata as factors, baseline measurements of the outcome, and the CARES as dependent variable. Exploratively, time span from baseline was additionally added to the MMRM as factor. A sensitivity analysis on imputed values complemented the analysis. Imputation was done using multiple reference based imputation followed by ANCOVA models.<sup>25</sup> Non-parametric Brunner-Munzel test for paired data (NBMP<sup>26</sup>) were used to compare the pre- and post-module scores within each trial arm. Differences from baseline were compared between trial arms using the Non-parametric Brunner-Munzel test for unpaired data (NBMU<sup>26</sup>). If discontinuation from study was due to recidivism, the participant receives the maximal value for the remaining measurements.

To analyze the effects of each individual module, secondary endpoints were analyzed using the NBMP comparing the pre- and post-module scores within each trial arm. Additionally, differences from baseline were compared between the trial arms using NBMU. Secondary endpoints were further analyzed by fitting linear models with trial arm and randomization strata, and pre module scores as factors and post module scores as dependent variable. If the assumptions of a linear model were not fulfilled either an ordinal regression model was fitted analogously instead or the score was dichotomized in order to fit a logistic regression model analogously. Analyses on secondary endpoints were carried out on all participants who provided pre- and post data for the respective module.

Frequencies and percentages of occurrences of re-offenses as reported by the SOs and of self-reported adverse events were compared using Fisher's exact tests. Ancilliary outcomes were not pre-specified. The WHO-5 was analyzed by MMRM with treatment group, time (six time-points), treatment-by-time interaction, and the randomization strata as factors, baseline measurements of the outcome and the WHO-5 as dependent variable. The same analysis rationale was applied for the WAI-SR COACH and WAI-SR SUPERVISOR with five time-points.

The error terms of all MMRMs were assumed to follow a multivariate normal distribution with unstructured covariance. Least squares mean changes from baseline are reported for

the trial arms with 95% confidence interval (CI) as well as the difference between the least squares treatment group means with 95% CI and p-value testing the null hypothesis of no treatment effect. For all multiple comparisons, p-values were adjusted using the Bonferroni-Holm’s method. All tests were performed two-sided with 5% significance level.

### 2.6. Role of funding source

The work was funded by the BMBF (Funding number: 01KR1807A). The funder had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

## 3. Results

Between March 01, 2020, and September 30, 2024, 385 individuals who committed sexual offenses against children have been screened for eligibility, of whom 16 have not been randomized (see Figure 1). 369 individuals were randomly assigned (182 placebo arm; 187 intervention arm). 213 participants have been included in the ITT sample (107 [57 · 2%] intervention arm; 106 [58 · 2%] placebo arm). 99 participants finished the WBI (44 [30 · 2%] intervention arm; 55 [22 · 4%] placebo arm). The trial was closed at September 30, 2024. 70 participants were not able to finish the WBI before closing (39 [59%] intervention arm; 31 [53%] placebo arm).

The trial profile is shown in Table 2. Only two screened participants gave informed consent to fill in the QNP, not allowing any inferences. Participants mean age was 41 years (range 20 to 68), predominantly male (216 [98%]; female: 2 [0 · 9%]; transsexual: 1 [0 · 5%]; other: 1 [0 · 5%]), and with a mean of 13 years of education (range 4 to 21 years). With regard to strata, 89 (40%) participants were mixed offenders (at least one conviction §176ff StG) and 132 (60%) hands-off offenders (only convictions based on §184b StGB). 176 (80%) participants were under CS (§§56, 57 StGB), 45 (20%) under post-release supervision (§§68ff StGB). 70 participants (32%) received an external psychotherapy and/or medical treatment, 151 (68%) not. The court ordered treatment for 124 (56%), for 87 (44%) not. 48 (39%) of all participants, who had an court ordered treatment, received external treatment, 76 (61%) not. 180 participants (81%) had a low to medium static recidivism risk, 41 (19%) a medium to high static risk to re-offend at baseline. The most common psychiatric conditions were other depressive syndromes (20 [9%]), and somatoform syndromes (17 [7 · 7%]). Psychosocial functioning was generally good (130 [59%] reporting no difficulty at all). Seven (3 · 2%) participants full-filled the diagnostic criteria for a pedophilic disorder, 213 (97%) not. Data on ethnicity was not assessed.

The mixed model for repeated measures (MMRM) for the primary endpoint showed no evidence of a treatment effect over time (all  $p > 0.15$ ). Strata as factors had no significant effect (all  $p > 0.13$ ; see Table 3). Sensitivity analysis (all  $p > 0.169$ ; appendix Table 8) and a MMRM additionally adjusted for time since the beginning of module one revealed no significance (all  $p > 0.293$ ; appendix Table 11). Within-arm pairwise comparisons in the intervention arm showed a significant reduction at module one from median 12 (Q1 8 to Q3 18) to median 12 (Q1 6 to Q3 16,  $p=0.0050$ ,  $p_{\text{adj}}=0.030$ ) and at module 3 from median 11 (Q1 4 to Q3 17) to median 9 (Q1 5 to Q3 16,  $p<0.010$ ,  $p_{\text{adj}}=0.050$ ; appendix Table 9). In the placebo arm, there was a significant reduction at module 1 from median 11 (Q1 7 to Q3 16) to median 9 (Q1 4 to Q3 16,  $p<0.0001$ ,  $p_{\text{adj}}<0.0001$ ), at module two a increase from median 8 (Q1 4 to Q3 16) to median 10 (Q1 4 to Q3 14,  $p=0.050$ ,  $p_{\text{adj}}=0.21$ ), and a decrease at module three from median 10 (Q1 5 to Q3 18) to median 8 (Q1 5 to Q3 14,  $p=0.041$ ,  $p_{\text{adj}}=0.21$ ; appendix Table 10). Change from baseline between trial arms showed no significant effects (all  $p > 0.052$ ; appendix Table 11). The SAP-conform analysis of the IoD yielded no significant effects (all  $p > 0.588$ ; appendix section 6.2).

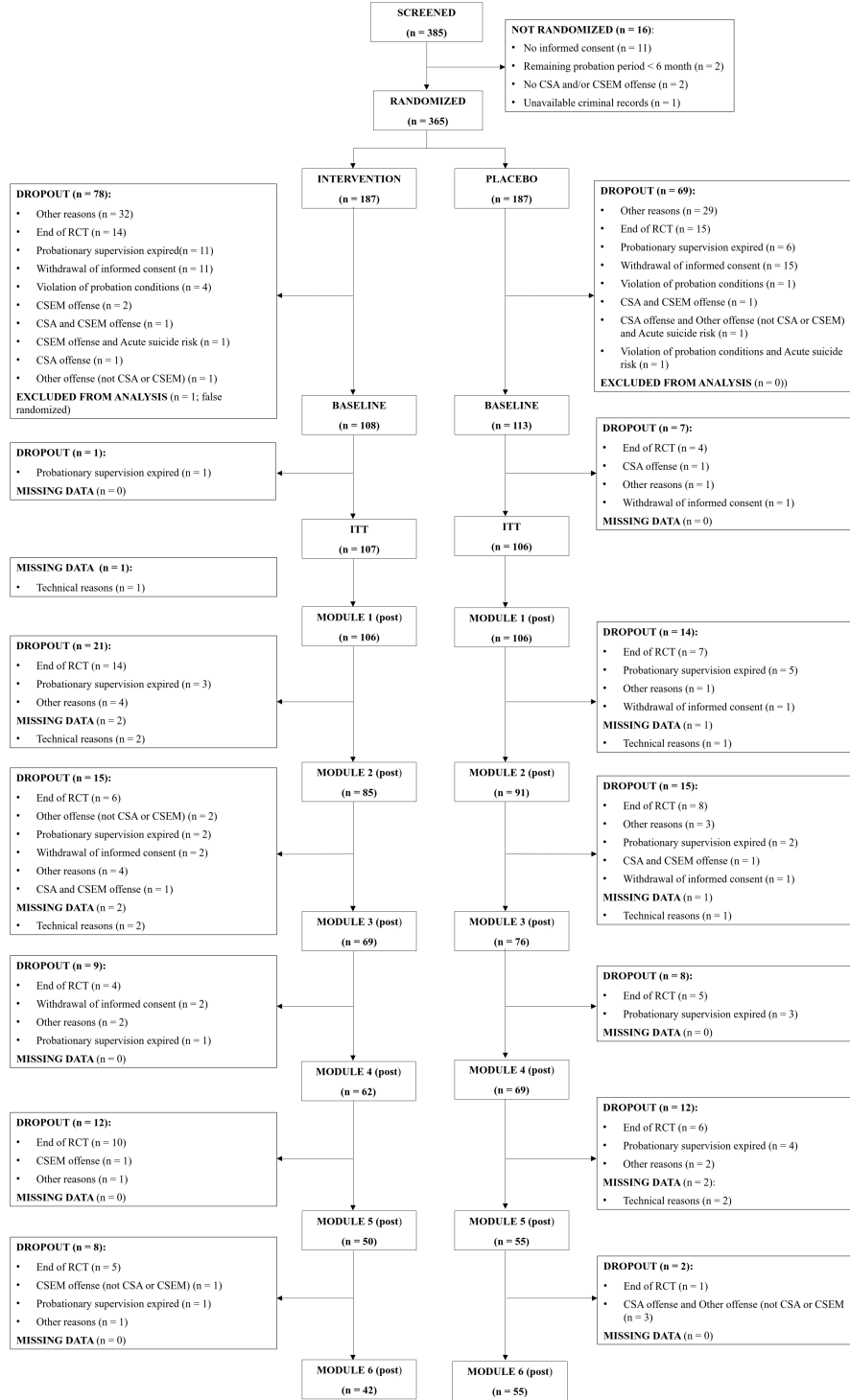


Figure 1: Trial profile. ITT = Intention-to-treat population. StGB = German penalty law.

Analysis of secondary outcomes showed for the intervention arm a significantly decrease in ESIQ from median 87 (Q1 80 to Q3 98) to median 87 (Q1 80 to Q3 90;  $p=0 \cdot 0010$ ,  $p_{\text{adj}}=0 \cdot 016$ ), indicating a reduction in sexual interests in children. The SSIC increased from median 28 (Q1 26 to Q3 30) to median 30 (Q1 25 to Q3 30,  $p=0 \cdot 049$ ,  $q=0 \cdot 59$ ), the CVTRQ increased from median 85 (Q1 79 to Q3 90) to median 87 (Q1 80 to Q3 92), indicating improved treatment readiness ( $p=0 \cdot 0060$ ;  $p_{\text{adj}}=0 \cdot 078$ ), and the DERS increased from median 5 (Q1 5 to Q3 6  $\cdot$  5) to median 6 (Q1 5 to Q3 8,  $p=0 \cdot 0050$ ,  $p_{\text{adj}}=0 \cdot 070$ ), indication deterioration of emotion regulation. All other secondary endpoints revealed not significance (all  $p \geq 0 \cdot 066$ ; Table 5). In the placebo arm, only the F-Soz-U increased from median 3  $\cdot$  86 (Q1 3  $\cdot$  14 to Q3 4  $\cdot$  43) at baseline to median 4 (Q1 3  $\cdot$  29 to Q3 4  $\cdot$  71) at post-measurement ( $p=0 \cdot 0030$ ,  $p_{\text{adj}}=0 \cdot 045$ ), no other significant differences were observed (all  $p \geq 0 \cdot 12$ ; Table 4).

Exploratory analyses of change from baseline showed that the ESIQ decreased in the intervention arm (median change  $-3$ , Q1  $-8$  to 0), but remained largely unchanged in the placebo arm (median change 0, Q1  $-1$  to Q3 0,  $p=0 \cdot 0020$ ,  $p_{\text{adj}}=0 \cdot 030$ ). Change from baseline in the DERS was greater in the intervention arm (median change 0, Q1 0 to 1) than in the placebo arm (median change 0, Q1 0 to Q3 0,  $p=0 \cdot 024$ ,  $p_{\text{adj}}=0 \cdot 34$ ). All other secondary endpoints showed no significant differences between trial arms (all  $p \geq 0 \cdot 077$ ; appendix Table 18). Explorative regression models fitting randomization strata as factors support this figure (appendix Section 7).

Table 2: Baseline characteristics by trial arm.

| Characteristic                                   | Overall N = 221 <sup>†</sup> | Intervention N = 108 <sup>†</sup> | Placebo N |
|--------------------------------------------------|------------------------------|-----------------------------------|-----------|
| <b>Age</b> <sup>2</sup>                          | 42 (20 - 68)                 | 41 (20 - 66)                      | 43 (21 -  |
| <b>Years of education</b> <sup>3</sup>           | 13 (4 - 21)                  | 12 (7 - 21)                       | 13 (4 - 2 |
| <b>Highest education</b> <sup>3</sup>            |                              |                                   |           |
| No formal education                              | 17 (7 · 7%)                  | 10 (9 · 3%)                       | 7 (6 · 2% |
| Lower secondary school                           | 88 (40%)                     | 42 (39%)                          | 46 (41%   |
| Intermediate secondary school                    | 70 (32%)                     | 36 (33%)                          | 34 (30%   |
| Upper secondary school                           | 30 (14%)                     | 13 (12%)                          | 17 (15%   |
| University degree                                | 16 (7 · 2%)                  | 7 (6 · 5%)                        | 9 (8 · 0% |
| <b>Employment status</b> <sup>3</sup>            |                              |                                   |           |
| Unemployed                                       | 61 (28%)                     | 29 (27%)                          | 32 (28%   |
| In education/training                            | 11 (5 · 0%)                  | 6 (5 · 6%)                        | 5 (4 · 4% |
| Unskilled worker                                 | 8 (3 · 6%)                   | 5 (4 · 6%)                        | 3 (2 · 7% |
| Worker                                           | 61 (28%)                     | 32 (30%)                          | 29 (26%   |
| Employee                                         | 51 (23%)                     | 25 (23%)                          | 26 (23%   |
| Self-employed                                    | 8 (3 · 6%)                   | 4 (3 · 7%)                        | 4 (3 · 5% |
| Retired                                          | 21 (9 · 5%)                  | 7 (6 · 5%)                        | 14 (12%   |
| <b>Marital status</b> <sup>3</sup>               |                              |                                   |           |
| Single                                           | 144 (65%)                    | 79 (73%)                          | 65 (58%   |
| Married                                          | 44 (20%)                     | 15 (14%)                          | 29 (26%   |
| Divorced                                         | 29 (13%)                     | 12 (11%)                          | 17 (15%   |
| Widowed                                          | 4 (1 · 8%)                   | 2 (1 · 9%)                        | 2 (1 · 8% |
| <b>Gender</b> <sup>4</sup>                       |                              |                                   |           |
| Female                                           | 2 (0 · 9%)                   | 2 (1 · 9%)                        | 0 (0%     |
| Male                                             | 216 (98%)                    | 105 (97%)                         | 111 (99%  |
| Trans/Transsexual                                | 1 (0 · 5%)                   | 0 (0%)                            | 1 (0 · 9% |
| Other                                            | 1 (0 · 5%)                   | 1 (0 · 9%)                        | 0 (0%     |
| Missing                                          | 1                            | 0                                 | 1         |
| <b>Sexual orientation</b> <sup>4</sup>           |                              |                                   |           |
| Exclusively heterosexual                         | 145 (66%)                    | 72 (67%)                          | 73 (65%   |
| Predominantly heterosexual                       | 24 (11%)                     | 10 (9 · 3%)                       | 14 (13%   |
| Bisexual                                         | 26 (12%)                     | 12 (11%)                          | 14 (13%   |
| Predominantly homosexual                         | 3 (1 · 4%)                   | 2 (1 · 9%)                        | 1 (0 · 9% |
| Exclusively homosexual                           | 11 (5 · 0%)                  | 5 (4 · 6%)                        | 6 (5 · 4% |
| Asexual                                          | 3 (1 · 4%)                   | 3 (2 · 8%)                        | 0 (0%     |
| Other                                            | 8 (3 · 6%)                   | 4 (3 · 7%)                        | 4 (3 · 6% |
| Missing                                          | 1                            | 0                                 | 1         |
| <b>Offense type</b> <sup>2</sup>                 |                              |                                   |           |
| Hands-on (at least one conviction §176ff StGB)   | 89 (40%)                     | 44 (41%)                          | 45 (40%   |
| Hands-off (only §184b StGB)                      | 132 (60%)                    | 64 (59%)                          | 68 (60%   |
| <b>Type of supervision</b> <sup>5</sup>          |                              |                                   |           |
| post-release supervision                         | 45 (20%)                     | 20 (19%)                          | 25 (22%   |
| community supervision                            | 176 (80%)                    | 88 (81%)                          | 88 (78%   |
| <b>Static recidivism risk</b> <sup>6</sup>       |                              |                                   |           |
| Low-medium                                       | 180 (81%)                    | 87 (81%)                          | 93 (82%   |
| Medium-high risk                                 | 41 (19%)                     | 21 (19%)                          | 20 (18%   |
| <b>Term of imprisonment (month)</b> <sup>7</sup> | 22 (0 - 156)                 | 21 (0 - 126)                      | 23 (0 - 1 |
| Missing                                          | 3                            | 1                                 | 2         |
| <b>Treatment ordered by court</b> <sup>7</sup>   |                              |                                   |           |
| Yes                                              | 124 (56%)                    | 62 (57%)                          | 62 (55%   |
| No                                               | 97 (44%)                     | 46 (43%)                          | 51 (45%   |
| <b>Additional treatment</b> <sup>2</sup>         |                              |                                   |           |
| Yes                                              | 70 (32%)                     | 33 (31%)                          | 37 (33%   |
| No                                               | 151 (68%)                    | 75 (69%)                          | 76 (67%   |
| <b>Major depressive syndrome</b> <sup>8</sup>    |                              |                                   |           |
| Yes                                              | 12 (5 · 4%)                  | 6 (5 · 6%)                        | 6 (5 · 3% |
| No                                               | 209 (95%)                    | 102 (94%)                         | 107 (95%  |
| <b>Other depressive syndromes</b> <sup>8</sup>   |                              |                                   |           |
| Yes                                              | 20 (9 · 0%)                  | 12 (11%)                          | 8 (7 · 1% |
| No                                               | 201 (91%)                    | 96 (89%)                          | 105 (93%  |
| <b>Panic syndrome</b> <sup>8</sup>               |                              |                                   |           |
| Yes                                              | 5 (2 · 2%)                   | 2 (1 · 0%)                        | 2 (2 · 1% |

Table 3: Mixed model for repeated measures (MMRM) for the primary endpoint Child Sexual Abuse Risk Evaluation Self-Report (CARES total score).

| Variable                                        | Beta  | 95% CI        | p-value |
|-------------------------------------------------|-------|---------------|---------|
| <b>(Intercept)</b>                              | -0.34 | -1.9 to 1.2   | 0.67    |
| <b>Type of supervision</b>                      |       |               |         |
| community supervision                           | 0.00  | Ref.          |         |
| post-release supervision                        | 0.32  | -1.2 to 1.8   | 0.68    |
| <b>Additional treatment</b>                     |       |               |         |
| No                                              | 0.00  | Ref.          |         |
| Yes                                             | -0.92 | -2.1 to 0.27  | 0.13    |
| <b>Offense type</b>                             |       |               |         |
| Hands-off (only §184b StGB)                     | 0.00  | Ref.          |         |
| Hands-on (at least one conviction §176 ff StGB) | 0.93  | -0.29 to 2.2  | 0.13    |
| <b>Baseline value</b>                           | 0.84  | 0.77 to 0.91  | <0.0001 |
| <b>Static recidivism risk</b>                   | 0.38  | -0.13 to 0.88 | 0.14    |
| <b>Timepoint * treatment</b>                    |       |               |         |
| Module 1 (post) * Intervention                  | 0.39  | -0.83 to 1.6  | 0.53    |
| Module 2 (post) * Intervention                  | -0.97 | -2.6 to 0.65  | 0.24    |
| Module 3 (post) * Intervention                  | -0.33 | -1.9 to 1.3   | 0.68    |
| Module 4 (post) * Intervention                  | -0.25 | -1.8 to 1.3   | 0.75    |
| Module 5 (post) * Intervention                  | -0.14 | -2.0 to 1.8   | 0.89    |
| Module 6 (post) * Intervention                  | -0.61 | -2.6 to 1.4   | 0.54    |
| <b>Timepoint</b>                                |       |               |         |
| Module 1 (post)                                 | 0.00  | Ref.          |         |
| Module 2 (post)                                 | 0.98  | -0.01 to 2.0  | 0.052   |
| Module 3 (post)                                 | -0.46 | -1.4 to 0.50  | 0.34    |
| Module 4 (post)                                 | -0.53 | -1.6 to 0.51  | 0.31    |
| Module 5 (post)                                 | 0.07  | -1.1 to 1.3   | 0.91    |
| Module 6 (post)                                 | 0.16  | -0.97 to 1.3  | 0.78    |

Abbreviation: CI = Confidence Interval, StGB = German penalty law.

The MMRM for the cumulative time from baseline to finish module showed a significant treatment-by-time interaction across all timepoints (all  $p < 0.035$ , appendix Table 3). Post-hoc pairwise comparisons showed longer times in the intervention arm for all modules. Unadjusted analyses indicated significant differences at module four (mean 260 days, SD 138, intervention arm; mean 223 days, SD 122, placebo arm;  $p=0.048$ ) and module five (mean 325 days, SD 161 intervention arm; mean 258 days, SD 113 placebo arm,  $p=0.028$ ), whereas differences at modules one, two, three, and six did not reach significance (all  $p > 0.053$ ). After adjustment for multiple comparisons, none remained significant (all  $q > 0.17$ ; appendix Table 4).

The MMRM for the time working on content revealed a significant treatment-by-time interaction at module one ( $\beta$  360 hours, 95% CI 49 to 671,  $p=0.024$ ), module four ( $\beta$  236 hours, 95% CI 19 to 453,  $p=0.034$ ), and at module five ( $\beta$  324 hours, 95% CI 6.1 to 641,  $p=0.046$ ). No significance was observed at module two ( $\beta$  148 hours, 95% CI -48 to 345,  $p=0.14$ ), module three ( $\beta$  188 hours, 95% CI -96 to 473,  $p=0.19$ ), and module six ( $\beta$  93 hours, 95% CI -142 to 328,  $p=0.42$ ; appendix Table 5). Post-hoc pairwise comparisons

reveled no differences between trial arms (all  $p \geq 0.18$ ; appendix Table 6)



During trial, no critical events were reported by SOs or STs. SOs excluded five (2.3%) participants due to recidivism (two [1.6%] intervention arm; three [2.7%] placebo arm; appendix Table 37). In the MRQ, preparation of contact with children, urges related to CSA, or urges related to CSEM were not reported. Two participants (0.9%) reported planning of a contact with a child (one [0.9%] intervention arm, one [0.9%] placebo arm). Two participants reported suicidal ideations (1.9%), two (1.9%) high sexual tension, and one participant (0.9%) reported losing control of sexual thoughts and activity in the intervention arm, none in the placebo arm. Feeling unable to cope with mental burden was reported by nine (4.1%) participants (7 [6.5%] intervention arm; two [1.8%] placebo arm). A current mental crisis was reported by 15 (6.8%) participants (11 [10%] intervention arm; four [3.5%] placebo arm), and very poor current mood by 12 (5.4%) participants (8 [7.4%] intervention arm; four [3.5%] placebo arm). However, none of these differences was statistically significant (all  $p \geq 0.062$ ; appendix Table 35).

For the WHO-5, the MMRM revealed no evidence for differences between trial arms over time, with no significant treatment-by-time interaction at any time-point (all  $p \geq 0.11$ ; appendix Table 38). Pairwise comparisons do not reach significance after correction for multiple testing (all  $p_{\text{adj}} \geq 0.14$ ).

The MMRM analysing the WAI-SR COACH showed significant treatment-by-time interaction effects at several timepoints ( $\beta$  2.4, 95% CI 0.30 to 4.5,  $p=0.025$  module one;  $\beta$  3.8, 95% CI 1.2 to 6.5,  $p=0.0044$  module two;  $\beta$  3.4, 95% CI 0.96 to 5.9,  $p=0.0067$  module three;  $\beta$  4.3, 95% CI 1.3 to 7.3,  $p=0.0051$  module five). At module four, there was no significant effect ( $\beta$  2.1, 95% CI -0.63 to 4.9,  $p=0.13$ ; appendix Table 42). Pairwise comparisons within the intervention arm showed significant increases at module one from median 43 (Q1 34 to Q3 49) to median 47 (Q1 39 to Q3 51,  $p=0.0020$ ,  $p_{\text{adj}}=0.010$ ), and at module five from median 47 (Q1 37 to Q3 57) to median 49 (Q1 41 to Q3 57,  $p=0.0040$ ,  $p_{\text{adj}}=0.016$ ). No other time-point showed significant pre-post changes (all  $p \geq 0.21$ ). In the placebo arm, no significant pre-post changes were observed at any timepoint (all  $p \geq 0.28$ ; appendix Tables 43-44). The change from baseline was greater in the intervention arm at module one (median 3, Q1 -2 to Q3 8 intervention arm; median 1, Q1 -3 to Q3 5 placebo arm,  $p=0.025$ ,  $p_{\text{adj}}=0.13$ ) and module five (median 2, Q1 0 to Q3 6 intervention arm, median 0, Q1 -2 to Q3 4 placebo arm,  $p=0.036$ ,  $p_{\text{adj}}=0.14$ ). No other comparison reached significance (all  $p \geq 0.059$ ; appendix Table 45).

The MMRM analysing the WAI-SR SUPERVISOR showed significant treatment-by-time interaction effects at four time-points, except module four ( $\beta$  2.3, 95% CI 0.64 to 4.1,  $p=0.0073$  module one;  $\beta$  3.0, 95% CI 0.87 to 5.2,  $p=0.0061$  module two;  $\beta$  3.2, 95% CI 1.2 to 5.3,  $p=0.0022$  module three; mean estimate 2.1, 95% CI -0.16 to 4.3,  $p=0.068$  module four;  $\beta$  3.6, 95% CI 1.2 to 6.0,  $p=0.0038$  module five; appendix Table 46). Pairwise comparisons within the intervention arm showed significant increases from median 44 (Q1 36 to 49) to median 47 (Q1 42 to Q3 53,  $p < 0.0001$ ,  $p_{\text{adj}} < 0.0001$ ) at module one and from 49 (Q1 ) to 50 (SD 8;  $p=0.046$ ,  $p_{\text{adj}}=0.18$ ). No other modules in the intervention arm showed significant pre-post changes (all  $p \geq 0.44$ ). In the placebo arm, no significant pre-post changes were observed at any timepoint (all  $p \geq 0.21$ ). At module one, change from baseline was higher in the intervention arm (median 4, Q1 -1 to Q3 6) than in the placebo arm (median 1, Q1 -3 to Q3 4,  $p=0.0090$ ,  $p_{\text{adj}}=0.045$ ). Similarly, change from baseline at module five was greater in the intervention arm (median 1, Q1 0

Table 4: Pairwise comparisons between pre- and post module scores of secondary endpoints within the placebo arm.

| Characteristic                                                   | Pre N = 113                | Post N = 113               |
|------------------------------------------------------------------|----------------------------|----------------------------|
| <b>CVTRQ (total score)</b>                                       |                            |                            |
| Mean (SD)                                                        | 85 (7)                     | 86 (8)                     |
| Median (Q1, Q3)                                                  | 86 (81, 91)                | 87 (81, 91)                |
| Min, Max                                                         | 67, 100                    | 61, 100                    |
| <b>RCQ (total score)</b>                                         |                            |                            |
| Mean (SD)                                                        | 25 · 7 (3 · 8)             | 26 · 1 (3 · 5)             |
| Median (Q1, Q3)                                                  | 26 · 0 (24 · 0, 28 · 0)    | 26 · 0 (24 · 0, 28 · 0)    |
| Min, Max                                                         | 16 · 0, 38 · 0             | 15 · 0, 32 · 0             |
| <b>F-Soz-U (total score)</b>                                     |                            |                            |
| Mean (SD)                                                        | 3 · 74 (0 · 84)            | 3 · 88 (0 · 89)            |
| Median (Q1, Q3)                                                  | 3 · 86 (3 · 14, 4 · 43)    | 4 · 00 (3 · 29, 4 · 71)    |
| Min, Max                                                         | 1 · 71, 5 · 00             | 1 · 71, 5 · 00             |
| Missing                                                          | 19                         | 19                         |
| <b>OQMPR (total score)</b>                                       |                            |                            |
| Mean (SD)                                                        | 29 (7)                     | 28 (8)                     |
| Median (Q1, Q3)                                                  | 29 (24, 34)                | 28 (22, 33)                |
| Min, Max                                                         | 15, 43                     | 12, 47                     |
| Missing                                                          | 19                         | 19                         |
| <b>UCLA (total score)</b>                                        |                            |                            |
| Mean (SD)                                                        | 28 · 2 (3 · 8)             | 28 · 1 (4 · 7)             |
| Median (Q1, Q3)                                                  | 27 · 0 (25 · 0, 30 · 0)    | 27 · 0 (25 · 0, 31 · 0)    |
| Min, Max                                                         | 20 · 0, 39 · 0             | 12 · 0, 42 · 0             |
| Missing                                                          | 19                         | 19                         |
| <b>BIS-15 (total score)</b>                                      |                            |                            |
| Mean (SD)                                                        | 29 (6)                     | 29 (6)                     |
| Median (Q1, Q3)                                                  | 28 (25, 32)                | 28 (25, 32)                |
| Min, Max                                                         | 16, 50                     | 17, 47                     |
| Missing                                                          | 35                         | 35                         |
| <b>CUSI (total score)</b>                                        |                            |                            |
| Mean (SD)                                                        | 27 (8)                     | 28 (8)                     |
| Median (Q1, Q3)                                                  | 26 (22, 30)                | 28 (23, 32)                |
| Min, Max                                                         | 16, 60                     | 16, 59                     |
| Missing                                                          | 35                         | 35                         |
| <b>DERS (subscale impulsivity)</b>                               |                            |                            |
| Mean (SD)                                                        | 6 · 08 (1 · 75)            | 5 · 92 (1 · 55)            |
| Median (Q1, Q3)                                                  | 5 · 00 (5 · 00, 7 · 00)    | 5 · 00 (5 · 00, 6 · 00)    |
| Min, Max                                                         | 5 · 00, 13 · 00            | 5 · 00, 10 · 00            |
| Missing                                                          | 35                         | 35                         |
| <b>NARQ (subscale externalizing strategies)</b>                  |                            |                            |
| Mean (SD)                                                        | 1 · 88 (2 · 75)            | 1 · 59 (2 · 12)            |
| Median (Q1, Q3)                                                  | 1 · 00 (0 · 00, 3 · 00)    | 1 · 00 (0 · 00, 2 · 00)    |
| Min, Max                                                         | 0 · 00, 16 · 00            | 0 · 00, 11 · 00            |
| Missing                                                          | 35                         | 35                         |
| <b>SPSI-R (total score)</b>                                      |                            |                            |
| Mean (SD)                                                        | 13 · 54 (2 · 56)           | 13 · 56 (2 · 78)           |
| Median (Q1, Q3)                                                  | 13 · 70 (12 · 10, 15 · 20) | 13 · 80 (11 · 90, 15 · 20) |
| Min, Max                                                         | 6 · 40, 20 · 00            | 6 · 40, 20 · 00            |
| Missing                                                          | 41                         | 41                         |
| <b>BMS (total score)</b>                                         |                            |                            |
| Mean (SD)                                                        | 49 (14)                    | 48 (14)                    |
| Median (Q1, Q3)                                                  | 43 (38, 54)                | 42 (39, 53)                |
| Min, Max                                                         | 38, 102                    | 38, 102                    |
| Missing                                                          | 54                         | 54                         |
| <b>ESIQ (total score)</b>                                        |                            |                            |
| Mean (SD)                                                        | 82 (16)                    | 82 (17)                    |
| Median (Q1, Q3)                                                  | 82 (80, 90)                | 81 (77, 91)                |
| Min, Max                                                         | 46, 127                    | 48, 121                    |
| Missing                                                          | 60                         | 60                         |
| <b>SOI-R (subscale desire for sexual activity with children)</b> |                            |                            |
| Mean (SD)                                                        | 3 (7)                      | 4 (9)                      |

to Q3 4) than in the placebo arm (median 0, Q1  $-2$  to Q3 2,  $p=0.061$ ,  $p_{\text{adj}}=0.24$ ), but reached not significance after correction for multiple comparisons. No other comparison reached significance (all  $p \geq 0.40$ ).

#### 4. Discussion

In this randomized placebo-controlled trial of a WBI for ICCO, no superiority of the intervention arm compared to the placebo arm was observed. Among secondary outcomes, only most notably a reduced sexual interest in children was observed. Also in line with our hypothesis, treatment readiness was enhanced in the intervention arm. Against the hypothesis, difficulties in emotion regulations increased significantly in the intervention arm, whereas the perceived social support increased isolated in the placebo arm. Only in the intervention arm robust significant improvements in the working alliance with STs and SOs were observed. No evidence emerged for an increased risk of recidivism. The frequency of adverse events was generally low without significant differences between trial arms.

Sexual interest in children is considered as the most influential risk-factor for sexual recidivism.<sup>20</sup> A central controversy concerns the modifiability of sexual interests in children. While the current clinical paradigm assumes limited changeability, this remains an active scientific debate. Recent meta-analyses reported indications of treatment-related changes in sexual interest, although none of the included studies employed an adequate control group.<sup>27,28</sup> Against this background, the present trial provides evidence for a possible reduction of sexual interest in children by treatment, even compared to a matched control group. However, the magnitude of change was small, warranting cautious interpretation of the clinical significance. The improvements on the working alliance with STs are encouraging, since there is a strong empirical association between working alliance with SOs and recidivism.<sup>29</sup>

The overall null effect seems to be in line with recent meta-analyses considering in-face treatment programs showing that studies with high methodological quality tend to report smaller or null effects.<sup>6,5</sup> Nevertheless, the result contrasts to Lätth et al. who reported a small, but significant effect in favor of the intervention arm on reducing the amount of viewing CSEM.<sup>13</sup> Conceptually, the current WBI differs from the WBI in this study, mainly regarding the amount of risk-factors which are targeted, the intensity, and the studied sample, questioning comparability. Previous evidence suggests that higher treatment intensity is generally associated with larger effects<sup>5</sup>, yet this relationship did not translate into superior outcomes in the present WBI. Meta-analyses also indicate that treatment targeting high-risk individuals yield larger effects.<sup>6,5</sup> Also dropout rates were high, reports on adverse events were generally low in both RCTs. For the first time, our study provides evidence for ICCO. Several mechanisms may underlie the null findings on the primary outcome. Dropout may have been driven in part by participants' interest in receiving official documentation for court rather than financial compensation. Baseline static risk-levels were low, suggesting a violation of the need principle of the RNR model by not adapting the content for individual needs.<sup>30</sup> Only module five provided individualized treatment content, but failed to show a significant reduction of offense-supportive conditions. Lack of direct contact with participants during the intervention limited participation retention efforts, as reminders could only be communicated via SOs. In consequence, the study was underpowered and small effects remain possibly undetected. Also, psychological burden in the sample was low, which contrasts with previous literature reporting substantial psychological distress.<sup>31,14</sup> Self-reports are controversial discussed, recent evidence suggests that self-report data

Table 5: Pairwise comparisons between pre- and post module scores of secondary endpoints within the intervention arm. The table shows the number of participants included in the analysis, the number of participants with available data at the outcome timepoint as well as the estimated effect size and its precision (such as 95% confidence interval) for the brunner-munzel test for pairwise comparisons.

| Characteristic                                  | Pre N = 108             | Post N = 108            | p        |
|-------------------------------------------------|-------------------------|-------------------------|----------|
| <b>CVTRQ (total score)</b>                      |                         |                         | 0 · 0060 |
| Mean (SD)                                       | 84 (7)                  | 86 (8)                  |          |
| Median (Q1, Q3)                                 | 85 (79, 90)             | 87 (80, 92)             |          |
| Min, Max                                        | 63, 97                  | 64, 100                 |          |
| <b>RCQ (total score)</b>                        |                         |                         | 0 · 38   |
| Mean (SD)                                       | 25 · 7 (4 · 0)          | 25 · 2 (4 · 0)          |          |
| Median (Q1, Q3)                                 | 26 · 0 (23 · 0, 28 · 5) | 25 · 0 (23 · 5, 28 · 0) |          |
| Min, Max                                        | 15 · 0, 33 · 0          | 12 · 0, 33 · 0          |          |
| <b>F-Soz-U (total score)</b>                    |                         |                         | 0 · 35   |
| Mean (SD)                                       | 3 · 83 (0 · 92)         | 3 · 85 (1 · 04)         |          |
| Median (Q1, Q3)                                 | 4 · 00 (3 · 14, 4 · 57) | 4 · 00 (3 · 29, 4 · 86) |          |
| Min, Max                                        | 1 · 00, 5 · 00          | 1 · 00, 5 · 00          |          |
| Missing                                         | 17                      | 17                      |          |
| <b>OQMPR (total score)</b>                      |                         |                         | 0 · 084  |
| Mean (SD)                                       | 29 (9)                  | 28 (9)                  |          |
| Median (Q1, Q3)                                 | 29 (23, 35)             | 29 (21, 35)             |          |
| Min, Max                                        | 12, 52                  | 12, 47                  |          |
| Missing                                         | 17                      | 17                      |          |
| <b>UCLA (total score)</b>                       |                         |                         | 0 · 066  |
| Mean (SD)                                       | 28 · 5 (3 · 9)          | 28 · 0 (4 · 1)          |          |
| Median (Q1, Q3)                                 | 28 · 0 (25 · 0, 31 · 0) | 27 · 0 (24 · 0, 31 · 0) |          |
| Min, Max                                        | 22 · 0, 37 · 0          | 21 · 0, 36 · 0          |          |
| Missing                                         | 17                      | 17                      |          |
| <b>BIS-15 (total score)</b>                     |                         |                         | 0 · 69   |
| Mean (SD)                                       | 30 (7)                  | 30 (8)                  |          |
| Median (Q1, Q3)                                 | 30 (24, 34)             | 29 (24, 35)             |          |
| Min, Max                                        | 15, 48                  | 15, 48                  |          |
| Missing                                         | 36                      | 36                      |          |
| <b>CUSI (total score)</b>                       |                         |                         | 0 · 62   |
| Mean (SD)                                       | 29 (9)                  | 29 (10)                 |          |
| Median (Q1, Q3)                                 | 28 (25, 32)             | 28 (21, 35)             |          |
| Min, Max                                        | 16, 66                  | 16, 63                  |          |
| Missing                                         | 36                      | 36                      |          |
| <b>DERs (subscale impulsivity)</b>              |                         |                         | 0 · 0050 |
| Mean (SD)                                       | 6 · 24 (2 · 26)         | 6 · 72 (2 · 25)         |          |
| Median (Q1, Q3)                                 | 5 · 00 (5 · 00, 6 · 50) | 6 · 00 (5 · 00, 8 · 00) |          |
| Min, Max                                        | 5 · 00, 17 · 00         | 5 · 00, 15 · 00         |          |
| Missing                                         | 36                      | 36                      |          |
| <b>NARQ (subscale externalizing strategies)</b> |                         |                         | 0 · 13   |
| Mean (SD)                                       | 1 · 94 (2 · 74)         | 2 · 11 (2 · 50)         |          |
| Median (Q1, Q3)                                 | 1 · 00 (0 · 00, 3 · 00) | 1 · 00 (0 · 00, 3 · 50) |          |
| Min, Max                                        | 0 · 00, 14 · 00         | 0 · 00, 11 · 00         |          |
| Missing                                         | 36                      | 36                      |          |
| <b>SPSI-R (total score)</b>                     |                         |                         | 0 · 93   |
| Mean (SD)                                       | 13 · 4 (3 · 2)          | 13 · 5 (3 · 3)          |          |
| Median (Q1, Q3)                                 | 13 · 8 (10 · 4, 16 · 0) | 13 · 5 (11 · 4, 15 · 6) |          |
| Min, Max                                        | 6 · 2, 19 · 8           | 7 · 4, 19 · 8           |          |
| Missing                                         | 46                      | 46                      |          |
| <b>BMS (total score)</b>                        |                         |                         | 0 · 35   |
| Mean (SD)                                       | 51 (14)                 | 53 (16)                 |          |
| Median (Q1, Q3)                                 | 46 (40, 57)             | 47 (39, 63)             |          |
| Min, Max                                        | 38, 86                  | 38, 99                  |          |
| Missing                                         | 55                      | 55                      |          |
| <b>ESIQ (total score)</b>                       |                         |                         | 0 · 0010 |
| Mean (SD)                                       | 90 (12)                 | 87 (12)                 |          |
| Median (Q1, Q3)                                 | 87 (80, 98)             | 85 (80, 90)             |          |
| Min, Max                                        | 66, 127                 | 66, 127                 |          |
| Missing                                         | 66                      | 66                      |          |

can be informative regarding recidivism risk.<sup>32</sup> Nevertheless, the CARES is a new instrument, and data linking it to official recidivism outcomes are not yet available.<sup>21</sup> Finally, participants in the in both arms required more time to complete modules than advised. Although differences were attenuated after correction for multiple testing, this prolonged duration may have contributed to dropout.

A representative sample was not reached due to data protection regulations. Financial compensation was provided to reduce a potential volunteer bias, but selective referral by probation officers cannot be ruled out. Also, engaging with the WBI necessitated an ability to read German, which is not available to everyone. At the same time, to our knowledge, this is the first RCT evaluating the effectiveness of a WBI for ICCO with a comparable high sample size targeting all meaningful risk-factors with one of the highest sample sizes reported in the literatur.

Future research should include in-person treatment comparators to clarify the relative effectiveness of the WBI. A stricter RNR-based individualization, the integration into blended approaches, and a condensation of the content may enhance adherence and effectiveness. Caution remains against the application of WBIs for ICCO at the moment as the observed deterioration of emotion regulation underlines the necessity of rigorous evaluations prior to implementation.

## Contributors

PF: writing – original draft, visualization, conceptualization, funding acquisition, project administration, supervision, software, methodology. SSch: writing – review & editing, investigation, methodology, resources, project administration. BS: writing – review & editing, investigation, resources, supervision. ST: writing – review & editing, investigation, conceptualization, resources. AL: formal analysis, visualization, writing – review & editing. FK: formal analysis, visualization, writing – review & editing. LM: writing – review & editing, investigation, methodology, resources. SH: writing – review & editing, conceptualization. CB: writing – review & editing, conceptualization, funding acquisition, resources. KJ: writing – review & editing, project administration. SE: writing – review & editing, conceptualization, funding acquisition. MR: writing – review & editing, conceptualization, funding acquisition. AS: writing – review & editing, conceptualization, funding acquisition. PB: writing – review & editing, conceptualization, funding acquisition. JLM: writing – review & editing, conceptualization, funding acquisition.

## Data sharing

The source code, the content of the WBI, and the de-identified data will be made available on reasonable request from the corresponding author (peter.fromberger@med.uni-goettingen.de). Requests, which can be submitted via email to the corresponding author, must include a detailed description of the research question and aims. Data sharing will be authorized following proposal approval and the execution of a signed data access agreement, strictly for scientific use. Code and content sharing follows the same procedure, strictly for non-commercial use.

## Declaration of interests

The authors declare no competing interests.

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